

Spacewalk-18: A Benchmark for Multimodal and Long-form Procedural Video Understanding in Novel Domains

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Abstract

Learning from (procedural) videos has increasingly served as a pathway for embodied agents to acquire skills from human demonstrations. To do this, video understanding models must be able to obtain structured understandings, such as the temporal segmentation of a demonstration into sequences of actions and skills, and to generalize the understandings to novel environments, tasks, and problem domains. In pursuit of this goal, we introduce Spacewalk-18, a benchmark containing two tasks: (1) step recognition and (2) video question answering, over a dataset of temporally segmented and labeled tasks in International Space Station spacewalk recordings. In tandem, the two tasks quantify a model's ability to: (1) generalize to novel domains; (2) utilize long temporal context and multimodal (e.g. visual and speech) information. Our extensive experimental analysis highlights the challenges of Spacewalk-18, but also suggests best practices for domain generalization and long-form understanding. Notably, we discover a promising adaptation via summarization technique that leads to significant performance improvement without model fine-tuning. The Spacewalk-18 benchmark is released at <https://brown-palm.github.io/Spacewalk-18>.

1. Introduction

This is Ground Control to Major Tom
 You've really made the grade
 And the papers want to know whose shirts you wear
 Now it's time to *leave the capsule*, if you dare

Space Oddity

Procedural videos, such as how-to or cooking, are produced to spread knowledge for human learners. With the recent advances in video understanding and robotic learning [3, 6, 48], these videos have become increasingly valuable learning resources for robots too. However, in order for machines to efficiently learn complex tasks and work alongside humans, they must be able to distill complex procedures into series of steps with only a few examples, often in a previously unseen environment. This challenge inspires



Figure 1. **Key properties of Spacewalk-18:** (1) Domain generalization: a side-by-side comparison of a sample frame from Spacewalk-18 and Ego4D [19] illustrates our benchmark's novel domain. (2) Multimodal: the visual content of the left frame does not align with its audio/speech. Instead, the speech corresponding to the right frame describes the left frame. (3) Long-form: the left frame shows the astronaut working on the solar array and the right frame shows him releasing bolts. These contextualize each other to identify that he is releasing the solar array in both frames.

the development of procedural video understanding systems that are able to generalize to previously unseen scenarios, and require minimal human supervision.

We introduce a new benchmark, **Spacewalk-18**, to advance multimodal, long-form, and procedural video understanding in a *novel* domain. Whereas most of the existing procedural video benchmarks are sourced from daily household scenarios [10, 19, 20, 26, 51, 53], the Spacewalk-18 dataset comprises video clips from 18 recorded extravehic-

ular activities (spacewalks) outside the International Space Station. Videos in this domain are naturally limited by the number of recorded spacewalks. As illustrated in Fig. 1, our dataset is first and foremost a benchmark to test pre-trained video-language models’ (e.g. [2, 61]) capability to *leave the capsule* and generalize to novel domains. Spacewalk-18 is also inherently multimodal and requires effective incorporation of long-form temporal context.

In Spacewalk-18, astronauts go on spacewalks for a variety of reasons, including to perform experiments, test equipment, or carry out maintenance/repairs. The recordings follow a fairly rigid agenda, often illustrated through a short animated sequence giving an overview of the steps planned for the spacewalk. Since our goal is to evaluate a (pre-trained) model’s generalization capability, we follow the design choice of recent benchmarks with similar goals (e.g., [34, 39]) and provide a moderate-sized training data for adaptation rather than pre-training. In order to collect detailed and dense temporal annotations for the training, validation, and test videos, we introduce a new protocol for temporal segmentation and action labeling to efficiently annotate the recordings and categorize content from the mission into the corresponding animated steps.

As illustrated in Figure 2, the collected annotations provide a structured representation of each spacewalk recording as a sequence of steps, each of which is in the form of a text description (e.g., “install thermal blanket on degraded antenna”), an animated illustration, and the temporal boundaries within the recording where the step is performed. We define two proxy tasks for evaluation:

1. The *step recognition* task evaluates the model’s ability to generalize to the spacewalk domain and incorporate video and text content into predictions.
2. The *question answering* task benchmarks the model’s ability to perform spatiotemporal reasoning.

We evaluate state-of-the-art video-language models, including contrastive video-language models (VLMs) [31, 61, 62, 67] and video large language models (VLLMs) [4, 9, 47, 59, 73, 76]. Through our experiments, we identify several best practices for incorporating multimodal information and long-form temporal context. Our human evaluation shows that an average English speaker achieves 67% step recognition performance after viewing 3.5-minute “training” example video, far exceeding any state-of-the-art VLLMs or proprietary APIs (e.g., GPT-5), suggesting a large room for improvement. Surprisingly, we discover that the most effective adaptation technique is to provide a frozen VLLM a summarized video as its context, and the relative gain is significantly larger compared to directly fine-tuning a model on the training split of Spacewalk-18.

Our contributions are three-fold: First, we propose a new procedural video benchmark on learning structured video representations and reasoning. Second, we collect the

Spacewalk-18 dataset with a new protocol for efficient annotation. It contains 96 hours of densely annotated videos and spans over 455 animated steps. Finally, we conduct extensive experimental analyses and discover the adaptation via summarization technique for effective domain generalization in Spacewalk-18.

2. Related Work

Procedural Video Understanding has important applications in video summarization [38], human machine interaction [37], procedural planning [8], and robotic learning [14, 65]. Common tasks defined for procedural video understanding include temporal action segmentation [13, 15] and detection [42], step localization [53, 77] and prediction [1, 18, 19]. They either use dense segment-level annotations or weak video-level labels [35, 43, 50]. Existing procedural video benchmarks are mainly sourced from two domains, how-to videos from online platforms [35, 49, 53, 77], and ego-centric videos collected from recruited actors [10, 19, 46] in kitchens and other household scenarios. More recently, the Perception Test [39] benchmark, aims to evaluate the zero-shot generalization capabilities of video-language foundation models. Our Spacewalk-18 benchmark is complementary to all of the existing procedural video understanding benchmarks: The spacewalk recordings are scripted and narrated in detail, yet cannot be solved by the visual or language modality alone. They exhibit strong structures and dependencies, and unfold over long temporal horizons. More importantly, unlike existing videos that are captured in kitchens and on earth, the space station is a novel domain, which simulates the real-world scenario of model deployment in unseen environments.

Video-language Foundation Models. Inspired by the success with large language models [7, 12], video-language foundation models have been proposed by training with large amounts of image and video data with masked token prediction [17] or video-task matching [74]. The videos are often accompanied by text descriptions such as speech narration [17, 52, 70, 71]. When video and language are encoded separately, a contrastive learning objective can be employed [31, 67] in a similar fashion as its image-based counterpart [40]. The objectives can be combined [69] and the encoders for different modalities can be shared [58]. In addition to joint multimodal pre-training, researchers have demonstrated the effectiveness of adapting visual descriptions into a large language model, such as with gated cross-attention [2], instructional tuning [9, 32, 73], linearly projecting the visual embeddings into the language space [29, 36], or augmenting large language models with video frame captions [59, 72]. Despite their amazing progress, we have shown that the state-of-the-art video-language models cannot generalize to Spacewalk-18 benchmark, whether zero-shot or with fine-tuning.

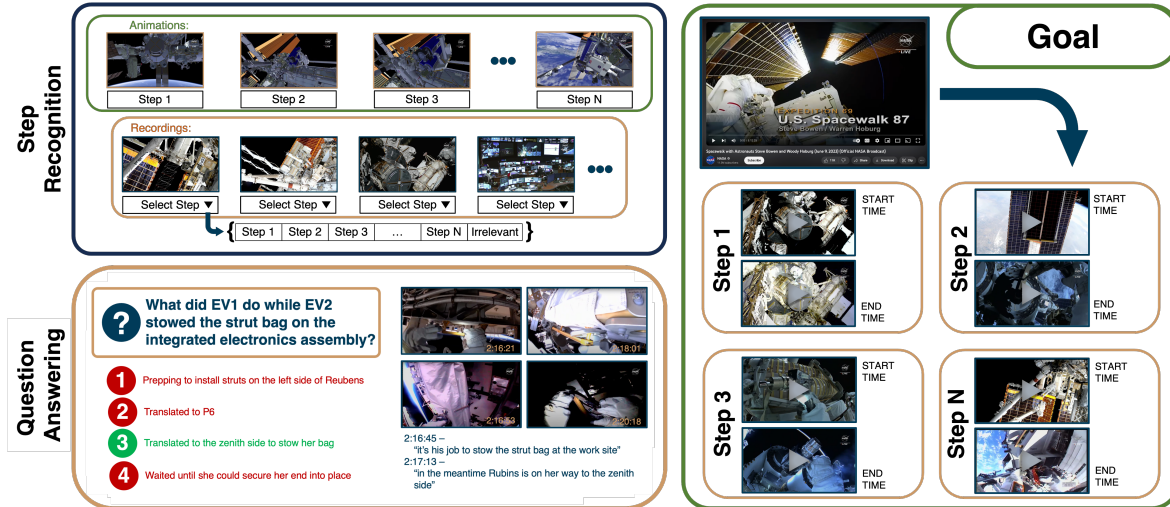


Figure 2. A spacewalk recording can be 7 or 8 hours long. The *step recognition* task aims to assign each video clip in the recording a step label, which is illustrated by a short animation and a text description. The *question answering* task targets video reasoning with long-term multimodal context. Both serve as intermediate benchmarks towards the “Goal”, which aims to represent a long procedural video as a sequence of steps and their corresponding video demonstrations for understanding and reasoning.

Long-form Video Understanding is an important open question for video representation learning. Earlier approaches [54] aim to extract high-level information such as character relationships from movies, where the solution is dominated by language-based approaches. Wu *et al.* [63] proposed an object-centric self-supervised framework, along with a benchmark for long-form video understanding. To encode long-form temporal context, memory-based approaches [64] have been proposed, along with new architectures that better scale with longer sequences [23, 30]. Our work is inspired by the recent dataset, EgoSchema [34], for question answering from long-form videos. Spacewalk-18 is complementary to EgoSchema, which is based on the same household videos as in Ego4D [19]. We also compute the temporal certificate [34] to measure the context needed for humans to solve a video understanding task, and observe that our task requires 40% more temporal context than EgoSchema.

3. The Spacewalk-18 Dataset

As shown in Figure 2, our goal is to segment spacewalk recordings into series of steps. A typical spacewalk video contains an animated preview of the steps to be performed, followed by the recording of astronauts performing the steps over multiple hours. We aim to annotate the spacewalk into a sequence of these steps. Each step corresponds to multiple video clips from recording, denoted by start and end times. Each step is accompanied with a text description annotated by a human expert, and illustrated by a video clip from the animated preview. We obtained 18 spacewalk recordings from YouTube. The total duration is 96 hours. For each video, we chunk the audio files into 10 minute clips and

feed them to Deepgram¹ to extract speech transcripts.

3.1. Labeling Process

Building our dataset requires temporally segmenting and labeling very long videos. Due to the presence of video clips irrelevant to the spacewalk (*e.g.*, scene of the mission control center in Figure 1), the temporal boundaries of a single step can be fragmented, making annotating the temporal boundaries time-consuming and measuring the inter-annotator agreement challenging. We propose to over-segment the recordings into short clips, each of which corresponds to at most a single step. We then ask the annotators to label each clip from a pre-defined list of steps for a spacewalk, while having access to an unbounded-length context from its neighboring video clips. The human annotators are thus free from having to draw detailed temporal boundaries, or to take multiple passes over the long video for step definition, temporal segmentation, and step labeling. We observe that this design drastically reduced human worker hours.

Define the Label Space. Our label space is derived from the animated preview for each spacewalk. We manually segment the animations into steps ourselves, and label each segment with a step caption. On average, this results in around 25 steps per spacewalk video.

Create Video Clips. We segment each video into short clips using PySceneDetect’s² shot boundary detection algorithm. We find that due to the long nature of the steps and tendency of the camera to switch angles often in spacewalk recordings, a given shot-segmented clip rarely spans over one step: we randomly inspect the annotated clips and

¹<https://deepgram.com>

²<https://github.com/Breakthrough/PySceneDetect>

observe that only 6% of them contain more than one step. Overall, only 4% of the frames are mislabeled due to shot boundaries. While imperfect, we believe the benefit of scaling up annotation significantly outweighs the minimal impacts on evaluation metrics.

Annotation Interface. To collect annotations, we build the Spacewalk Video Annotation Tool (example interface is shown in Appendix A.1). The interface displays the animated steps and clips for human annotators to label. They watch the annotation clips and select a label for each clip from the set of step labels, “Irrelevant”, or “Unsure”. This interface allows them to view the entire spacewalk recording as context. The “Irrelevant” label categorizes any clip that does not contain footage of one of the steps for the given spacewalk. This includes shots of the mission control center, noisy shots (*e.g.*, blue screen), and shots of “get-ahead” steps that were not originally planned for the spacewalk. We have at least three human workers annotate each clip and we choose the most commonly selected label as the true label. Our annotation tool is publicly released.

Merge Adjacent Clips. Since we intentionally over-segment the long spacewalk recordings before collecting annotations, we include a final step to use the collected annotations to obtain true temporal boundaries by merging all adjacent clips with identical labels.

We provide **supplementary videos** to illustrate the Spacewalk videos and our collected annotations.

3.2. Dataset Statistics, Diversity, and Difficulty

The 18 spacewalk videos are segmented into in total 31,812 clips for human annotation. After annotation, adjacent clips with the same step labels are merged as post-processing, resulting in 3,753 merged clips. These clips have an average length of 92 seconds and span over 96 hours in total. There are 455 animated step labels across the dataset (excluding “Irrelevant”). Each step takes on average 5 merged clips and has an average of 9 minutes of video. To analyze the diversity of the dataset, we filter the nouns and verbs from the step captions to obtain a list of objects and actions. Our dataset has 51 diverse objects (*e.g.* *battery*) and 47 atomic actions (*e.g.* *install* and *ingress*) (See Appendix A.3).

Spacewalk-18 is *open-vocabulary*, as the list of step labels varies across different videos. Example labels in the training set include “EV1 & EV2 remove the first battery” and “EV2 ingresses foot restraint”. In the test set, there are similar labels, such as “Chris & Bob remove new battery from slot A” and “Chris enters foot restraint”.

The visual content in Spacewalk-18 is out-of-domain for most existing models. The shots of astronauts working in space are visually distinct from any worldly actions. There is also a mix of first- and third-person point of view camera angles. Additionally, the two astronauts in each spacewalk often work on different steps in the same time, leading to

steps appearing in multiple non-continuous segments. To reduce the impact of task difficulty on annotation accuracy, we offer training to annotators from online platform, and those who achieve at least 80% accuracy on a held-out set after training are recruited to annotate the dataset.

3.3. Task Definition on Spacewalk-18

The 18 recordings in Spacewalk-18 contain varying numbers of steps. To ensure each split contains a balanced number of steps, we manually split the dataset into training, validation, and test sets in a ratio of 10:2:6. We introduce two multimodal long-form video understanding tasks on Spacewalk-18, step recognition and question answering:

Task 1 - Step Recognition. Our dataset divides the multi-hour spacewalk recording into several steps. Each step is described by an animated video clip, the transcript of its narration, and an annotated caption. Our step recognition task aims to recognize these steps from the spacewalk recording. Given a timestamp t and the list of K steps in the corresponding spacewalk mission, a model is tasked with determining the step occurring at timestamp t by predicting a label from $\{0, 1, 2, \dots, K\}$ (label 0 stands for “Irrelevant”). We notice that temporal contexts are crucial for recognizing the steps, from which we can see the astronauts’ actions and know the completed and remaining steps. However, current video-language models are incapable of digesting hours-long spacewalk recordings. So we set a context window length w and offer the video clip $[t - w/2, t + w/2]$ to the model. The clip includes both visual content and textual speech transcripts. We test models with varying w to investigate their ability to understand temporal contexts.

We construct 2000 samples from each training, validation, and test video, resulting in an average of 1 sample for every 10 seconds of video. As the visual content of spacewalk videos does not change rapidly, these samples are sufficient to represent an entire video. To balance the categories, $2000/(K + 1)$ timestamps are uniformly sampled from each step’s corresponding video clips. These timestamps are used as the middle timestamps t in the task definition above across different context lengths.

To evaluate model performance, we calculate Accuracy and mean Average Precision (mAP) to measure how accurately each sample is recognized. By merging temporally adjacent samples with the same predictions into intervals, we derive a temporal segmentation of each spacewalk mission. We also adopt the standard intersection-over-union (IoU) metric with the ground truth step boundary annotations to measure the segmentation correctness. We found that sampling more training or validation examples (thus denser in time) has little impact on the IoU metric, which are defined on continuous temporal boundaries. Detailed metric discussions can be found in Appendix B.

Task 2 - Question Answering. For ease of evaluating

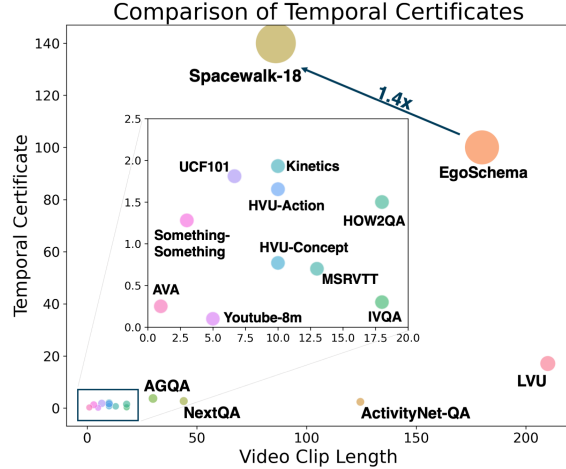


Figure 3. Temporal certificate (“long-form-ness”) lengths across commonly adopted datasets with action recognition and question answering annotations. Spacewalk-18 is 1.4x the length of the nearest comparable (EgoSchema). Figure adapted from [34].

video-language models on temporal understanding and reasoning with Spacewalk videos, we further introduce the question answering task. We first split each Spacewalk video into hour-long consecutive video segments, and then manually collect questions that are either high-level (e.g. *what is the goal of this mission*) or detailed and requires temporal localization and reasoning (e.g. *What type of equipment does the astronaut retrieve first, and how is it utilized during the mission*). Question answering is formulated as multiple-choice, where four candidates are provided. The questions and all possible choices are annotated by experienced annotators. To facilitate the question collection, we leverage the step annotations to automatically generate certain types of questions, such as “*What did EV1 do while EV2 did [task]*”. Overall, we collected 376 questions for testing a pre-trained VLM’s zero-shot performance.

3.4. Characteristics of Spacewalk-18

Our benchmark requires semantic understanding and temporal reasoning abilities in a truly *unique* domain. A comparable benchmark is the recently released Perception Test [39], which is also designed to measure similar skills and generalization capabilities. Spacewalk-18 nicely complements benchmarks like Perception Test as they focus on different domains (daily life vs. space) and has comparable total video durations (74 vs. 96 hours).

Long-form-ness: Spacewalk-18 not only contains long videos, but also requires high amounts of context (long-form video understanding) in order to annotate. We use the temporal certificate [34] metric to quantify the long-form-ness of our video dataset. The temporal certificate measures the amount of video context required for a given dataset/task. Human verifiers are provided with an annotated video clip and are asked to select the amount of the

Benchmarks	Duration (h)	# Annotations	Domain
Step recog. (task 1)		Temporal Segments	
Ikea-FA [56]	4	2 k	Assembling
Breakfast [26]	77	8 k	Cooking
YouCook2 [75]	176	14 k	Cooking
EPIC-KITCHENS [11]	100	89 k	Cooking
EgoPER [28]	28	8 k	Cooking
Assembly101 [46]	513	105 k	Assembling
EPIC-Tent [24]	5	1 k	Assembling
Spacewalk-18 (ours)	96	4 k	Spacewalk
Question ans. (task 2)		Questions	
EgoSchema [34]	250	5 k	Open-domain
Video-MME-L [16]	206	900	Open-domain
P. Test 1hr-walk [21]	10	70	Tours
Spacewalk-18 (ours)	96	376	Spacewalk

Table 1. Statistics comparison with single-domain step recognition benchmarks and recent long-form video QA benchmarks.

video they require to be confident that the provided label is correct. EgoSchema [34] defines benchmarks with certificate length around 1 second as short video tasks, 10 seconds as long-form, and 100 seconds as very long-form.

To collect this metric, we split each clip into 5-second chunks and task human workers with selecting the minimum subset of these clips that they need to be confident in the given label. In Figure 3, we benchmark our dataset with eight annotators over 2.5 hours of spacewalk video. Our dataset has an average clip length of 89 seconds and a temporal certificate length of 140 seconds. This is 1.4x the length of the nearest dataset and places it in the category of “very long-form video datasets” [34].

Multi-modality: We also observe that the temporal understanding and reasoning tasks in Spacewalk-18 are inherently multimodal. Table 3 shows the human performance when asked to solve the step recognition task, where they can freely explore the temporal context from the entire spacewalk video when needed. We can see that humans perform the best when both video and audio (speech transcripts) are available, with an accuracy of 67.0%. This multimodal performance is higher than when humans only have access to video (52.2%) or audio (39.1%).

Comparison with Relevant Datasets: Table 1 compares Spacewalk-18 with other single-domain step recognition datasets. Our dataset has comparable total duration as relevant datasets, while focusing on a unique domain than cooking or assembling furniture. We also compare with recent long-form video QA benchmarks, which shows that our question size is reasonable as a single-domain benchmark.

4. Recognition and Reasoning Models

For step recognition, we benchmark both contrastive video-language models (contrastive VLMs) and video large language models (VLLMs). Contrastive VLMs can provide aligned video and text embeddings while VLLMs can process video and language simultaneously and perform video question answering. Both of them are evaluated in zero-

shot scenario to demonstrate how pre-trained models generalize to Spacewalk-18. We also evaluate contrastive VLMs in fine-tuning scenario, which can effectively adapt pre-trained models to novel domains [44]. Both last-layer and all-layer fine-tuning are considered here.

In Spacewalk-18, the animation of the i -th step is described by an animation video V_i^a , a transcript of its narration T_i^a , and a step caption C_i^a . A spacewalk recording clip centered at timestamp t with length w contains a video clip $V_{t,w}$ and a transcript $T_{t,w}$. When evaluating contrastive VLMs, we format recognition as retrieval, similar to how contrastive VLMs can be used for zero-shot classification. We employ separate video encoder $F_v(\cdot)$ and text encoder $F_t(\cdot)$ to derive clip features, and match the spacewalk recording clips with step animations. To evaluate VLLMs, we feed the videos and transcripts into the models and treat our tasks as multi-choice video question answering. Unless otherwise mentioned, all the models use Sparse Frame Sampling to process videos, where we uniformly sample k frames from the entire clip $V_{t,w}$ regardless of the video length and feed them into the video encoder. k is the number of frames used during model pre-training, which may differ between different models. For the question answering task, a model is given the entire hour-long input video, and it needs to search for the relevant evidence from the long-form input, making the task more challenging.

4.1. Step Recognition and QA by VLLMs

Zero-shot. We format the step recognition task as multi-choice video question answering as the following. Given a spacewalk clip with K steps in the mission, we provide the video $V_{t,w}$ and transcript $T_{t,w}$ to the model and ask “Which step does the frame in the middle of this video belong to?” We also provide the caption C_i^a and transcript T_i^a of each step and require the model to choose a step index between 0 and K . See our prompt in Appendix C.2. We follow the same strategy for the question answering task, with the only difference that the choice candidates are provided by each question. We focus on the zero-shot setup and report performance not only on state-of-the-art VLLMs, but also proprietary APIs, such as GPT-4o.

4.2. Step Recognition by Contrastive VLMs

Zero-shot. To derive a feature for a step animation, we first extract the video, transcript, and caption features respectively, and then concatenate them. Formally, the feature of the i -th step is $f_i^a = [F_v(V_i^a), F_t(T_i^a), F_t(C_i^a)]$. To construct a feature for the “Irrelevant” category, we write a textual description of it and extract its text feature. The description is $DES = \text{“The mission control center, noisy shots (e.g. blue screen), or tasks not planned for the spacewalk.”}$ Hence the “Irrelevant” category feature is $f_0^a = [F_t(DES), F_t(DES), F_t(DES)]$. For a record-

ing clip, we concatenate the video and transcript features: $f^s = [F_v(V_{t,w}), F_t(T_{t,w}), F_t(T_{t,w})]$. Here, the transcript feature is repeated so that the feature dimensionality is the same as those of the animation features.

To form a prediction, we compute the similarities $f^s \cdot f_i^a$ between a recording clip and an animation step, and pick the step with the highest similarity.

Last-layer Fine-tuning. We fine-tune a linear layer upon the pre-trained models using the training set. Specifically, we freeze the models and train a linear layer $G_\theta(\cdot)$ mapping the spacewalk clip features to the animation features. We minimize the cross entropy loss during training, which is

$$\mathcal{L} = -\log \frac{\exp(G_\theta(f^s) \cdot f_y^a)}{\sum_{1 \leq i \leq K} \exp(G_\theta(f^s) \cdot f_i^a)}. \quad (1)$$

After training, instead of constructing or learning a feature for the “Irrelevant” category, we find a threshold τ and recognize all clips whose similarities to task steps are all below τ as “Irrelevant”. This is more reasonable than an “Irrelevant” feature because this category contains a subspace formed by various concepts rather than a single concept. With the threshold τ , we make predictions by

$$\hat{y} = \begin{cases} \arg \max_{1 \leq i \leq K} G_\theta(f^s) \cdot f_i^a, & \max G_\theta(f^s) \cdot f_i^a \geq \tau, \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

We find the τ with the best F1 score on the the validation set and use it in the test phase. In practice, this method performs better than constructing an “Irrelevant” feature through text descriptions.

All-layer Fine-tuning. We fine-tune the entire backbone of a pre-trained model to encode spacewalk video clips. We freeze the animation features and use the same cross entropy loss to fine-tune the model. The threshold τ for the “Irrelevant” category is also set using the validation set.

Incorporating Longer Temporal Context. Our Sparse Frame Sampling approach naturally incorporates long temporal context with the same computational cost, but may lose important details due to sampling. We therefore explore several alternatives to incorporate temporal context: **Dense Frame Sampling** at 1 FPS, which are directly encoded by the video encoder to obtain a single video embedding. This approach is bounded by GPU memory since the number of frames scales linearly with the context duration. **Long-term Feature Bank (LFB)** [64], which first divides the long context into seconds-long segments and leverages a frozen video encoder to extract one video embedding for each segment with a frame sampling rate of 1 FPS.

We explore the following LFB variants following [64]: Average pooling over the query and all of the context features to form a single embedding (**LFB Avg**); Average pooling the history context and future context separately, and concatenate them together with the query embedding (**LFB**

Cat); Learning to aggregate temporal context via non-local blocks [60] (LFB NL) or a two-layer Transformer encoder (LFB TF). More details can be found in Appendix C.5.

5. Experiments

Our experiments show that current video-language models perform significantly worse than humans on Spacewalk-18. We also demonstrate the importance of both vision and language on our tasks. Furthermore, the capability of LFB in incorporating temporal contexts is verified.

5.1. Evaluated Models

We evaluate contrastive VLMs and VLLMs on Spacewalk-18. For contrastive VLMs, we test **EgoVLP** [31], **VideoCLIP** [67], **InternVideo** [61], and **InternVideo2** [62]. For VLLMs, we test open-source **LLaVA-Next-Video** [73], **VideoLLaMA2** [9], **LongVU** [47], **Qwen2.5-VL** [4], and **InternVL3** [76], and proprietary **GPT-4o** and **GPT-5**. Moreover, recent works [59, 72] solve VideoQA tasks effectively by feeding generated video frame captions into LLMs. Hence, we develop a **caption-enhanced LLM** with LLaVA-1.5-13B [33] as the video frame captioner and GPT-4o as the LLM reasoner. See the selected checkpoints and number of sampled frames in Appendix C.1.

5.2. Human Performance Evaluation

On the step recognition task, we evaluate human performance with all our annotators. Each of them is assigned part of the video clips and has access to an unlimited video context. We measure the accuracy of each human performer and take their average as the human performance. Overall, they achieve an accuracy of 67%. This can be viewed as the “upperbound” performance on step recognition.

5.3. Main Results

Table 2 shows the model performances on Spacewalk-18. Due to the high computational cost of all-layer fine-tuning, it is only employed on InternVideo.

On the step recognition task, we report the results under a few context lengths and conduct a thorough exploration of it in Section 5.5. The best accuracy on this task is 25.75% (InternVL3-8B) for open-source models and 36.15% (GPT-5) for proprietary models. Their performances are far less than the human performance of 67%. This verifies that our task is reasonable for humans but challenging to models. VLLMs outperform contrastive VLMs, and significantly when the context window is long. Among the contrastive VLMs, InternVideo performs the best. Both last-layer and all-layer fine-tuning can boost contrastive VLMs, and all-layer fine-tuning yields higher performance than last-layer fine-tuning. However, the improvements are marginal, indicating that adapting models to extremely rare domains remains challenging.

Method	Step Recognition									QA
	$w = 1 \text{ min}$			$w = 3 \text{ min}$			$w = 5 \text{ min}$			
	Acc.	mAP	IoU	Acc.	mAP	IoU	Acc.	mAP	IoU	
Random	4.22	-	1.51	4.22	-	1.51	4.22	-	1.51	25.00
Human*	-	-	-	-	-	-	67.0	-	-	-
Zero-shot										
EgoVLP	7.18	8.28	1.90	7.71	9.08	1.67	7.59	9.83	1.59	-
VideoCLIP	8.00	10.24	2.12	6.58	11.11	2.65	7.85	10.68	2.38	-
InternVideo	9.35	10.78	2.96	9.48	11.01	2.73	9.07	11.56	2.97	-
InternVideo2	6.33	9.38	1.61	5.64	9.05	1.53	7.08	9.72	1.63	-
LLaVA-Next-Video-34B	10.35	-	3.77	13.71	-	5.07	13.82	-	4.92	29.52
VideoLLaMA2-7B	9.34	-	2.88	14.37	-	5.45	17.32	-	6.28	31.65
LongVU-7B	8.58	-	3.04	10.49	-	3.81	12.01	-	4.41	36.44
Qwen2.5-VL-7B	13.33	-	4.54	19.79	-	7.62	22.99	-	8.52	33.78
InternVL3-8B	16.07	-	6.19	21.72	-	8.83	25.75	-	10.79	31.65
Caption-enhanced LLM	18.56	-	9.33	26.32	-	12.89	28.49	-	13.16	30.37
GPT-4o	15.47	-	6.36	21.65	-	9.41	26.40	-	11.55	32.45
GPT-5	-	-	-	-	-	-	36.16	-	18.42	46.54
Last-layer Fine-tuning										
EgoVLP	6.34	8.92	2.17	9.68	10.66	3.03	10.21	10.86	3.18	-
VideoCLIP	8.40	9.80	3.21	9.98	11.14	3.71	8.87	10.39	3.46	-
InternVideo	10.12	12.17	4.02	11.13	12.68	4.04	10.08	12.53	4.04	-
InternVideo2	9.41	10.36	3.53	8.96	10.00	2.98	8.39	9.97	2.97	-
All-layer Fine-tuning										
InternVideo	13.21	11.77	4.60	13.34	12.93	4.54	12.93	13.09	4.63	-

Table 2. Model performances on Spacewalk-18. GPT-5 performs the best among all the models on both step recognition and question answering tasks. Both last-layer and all-layer fine-tuning improve the performances of contrastive VLMs. *: Humans have access to unlimited context.

Besides VLMs, we evaluate a video-only model, VideoMAE [55], on the step recognition task. Since the task is open-vocabulary, we train an MLP to project the VideoMAE embeddings to the SentenceBert [41] text embeddings of the step captions. It achieves 10.34% accuracy, 12.91% mAP, and 2.97% IoU with 5-minute context window length.

On the question answering task, GPT-5 shows strong performance (46.54%). The most performant open-source model is LongVU (36.44%), which has token compression mechanism designed for long-form video understanding. In contrast, all other models outperform the random baseline marginally. For both tasks, we provide **qualitative results** including success and failure cases in Appendix D.6.

5.4. Effect of Modality

In Table 3, we ablate the input modality on the step recognition task using zero-shot video-language models. And we also measure the human performances in multimodal and unimodal cases. In most cases, human and open-source video-language models perform the best when both modalities are provided, demonstrating the inherently multimodal nature of our task. However, the proprietary VLLMs achieve the highest accuracy when only text is given, probably due to their strong language prior. Besides, we notice that humans can better use the visual inputs, while our models benefit more from the texts. We hypothesize that humans excel in extracting abstract concepts from videos and match them with other modalities, while it is difficult for video-language models. Moreover, while the overall performance of InternVideo is worse than VLLMs, it outperforms most

Method	Accuracy					
	$w = 1 \text{ min}$			$w = 5 \text{ min}$		
	V	T	V+T	V	T	V+T
Human*	-	-	-	52.2	39.1	67.0
InternVideo	7.84	9.15	9.35	6.26	9.68	9.07
LLaVA-Next-Video	5.22	9.38	10.35	4.93	14.65	13.82
VideoLLaMA2	4.39	8.03	9.34	4.56	13.03	17.33
Qwen2.5-VL	6.59	13.71	13.33	6.98	22.78	22.99
Caption-enhanced LLM	5.48	18.92	18.56	5.13	31.03	28.49
GPT-4o	6.86	18.92	15.48	6.68	31.03	26.40
GPT-5	-	-	-	13.00	39.18	36.16

Table 3. Ablation about input modality on step recognition task. The models are evaluated in zero-shot scenario. V: video; T: text (captions and transcripts). *: Human has unlimited context and accesses transcripts in the form of audio.

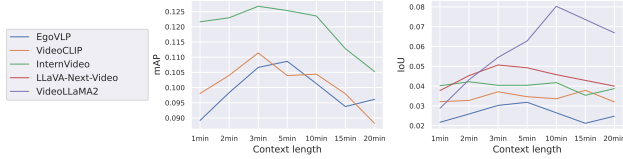


Figure 4. Ablation on context length. We test the models under various context lengths. Contrastive VLMs are last-layer fine-tuned while MLLMs are zero-shot. When the temporal context is extremely long, the models can no longer benefit from it.

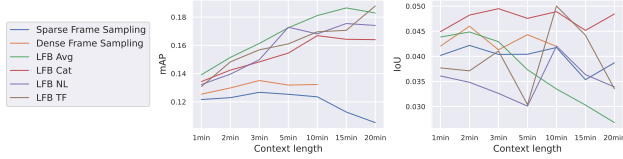


Figure 5. Performances of different temporal context incorporation methods built upon frozen InternVideo features. While LFB methods yield increasing mAP when the temporal context extends, one-time feed-forward models with either sparse or dense frame sampling cannot benefit from the context.

of the models in the video-only setting, demonstrating its better visual generalization to rare visual domains.

5.5. Leveraging Temporal Context

As shown in Figure 4, unlike humans that have temporal certificate of 2.3 minutes, the open-source video-language models cannot benefit from very long context on both tasks. As context window length increases, their performances initially improve, but start to decline after reaching their peaks.

To address this issue, we explore the temporal context incorporation approaches for contrastive VLMs introduced in Section 4.2 on the step recognition task. As Table 3 shows that contrastive VLMs have the most robust visual capability to rare domains, it is more effective to explore video context incorporation using them than VLLMs. The performance curves with respect to context lengths are shown in Figure 5. The two one-time feed-forward methods – Sparse/Dense Frame Sampling – have similar performances, and they are not improved given expanded temporal context windows as well. This indicates that naively sampling more frames cannot help pre-trained

Task	Method	Summary	Accuracy (%)
Step Recog.	InternVideo	-	9.48
	InternVideo (fine-tuned)	-	13.34
	Caption-enhanced LLM	-	23.50
	Caption-enhanced LLM	Step Oracle	28.49
QA	GPT-4o	-	32.45
	GPT-4o	Animation	55.59
	GPT-4o	Step Oracle	81.12

Table 4. Adaptation via summarization significantly improves Spacewalk-18 performance on both tasks.

video-language models to understand long-form videos. In contrast, the mAP curves of all the LFB methods show significant upward trends, verifying their ability to incorporate temporal contexts in long videos. Among them, LFB Cat, which concatenates past, present, and future video features, outperforms other methods on the accuracy and IoU metrics, demonstrating its effectiveness.

5.6. Adaptation via Summarization

We finally investigate how to effectively adapt a pre-trained model to solve Spacewalk-18. Our key inspiration is that for long-form, novel-domain videos, a model may adapt to and understand the videos by watching their summarizations.

In the step recognition task, the full list of steps is available to the VLLMs. This list in fact serves as an oracle summarizing the overall spacewalk mission, which might contribute to their better performance than contrastive VLMs. To ablate its influence, we give the caption-enhanced LLM only one step each time and ask it to rate each step from 1 to 10. In Table 4, the step oracle indeed improves its performance from 23.50% to 28.49%, which is an even higher gain than fine-tuning the InternVideo.

For the QA task, we simulate the summarization by either re-purposing the 5-minute animation video of each spacewalk, or by assuming a known list of steps that occur within the question video. These steps are represented by step captions and middle frames. In Table 4, the animation and step oracle provide significant improvements over GPT-4o. While both animation and step oracle should be considered as privileged information, they highlight the direction to adapt pre-trained models via *condensed* knowledge.

6. Conclusion

We introduce the Spacewalk-18 benchmark to evaluate video-language models’ capability to generalize to unseen domains, and their comprehension of multimodal information and long-term temporal context. We demonstrate that while average human annotators achieve competitive performance on our benchmark, existing video-language models still struggle with domain generalization and long-form video understanding. Meanwhile, we discover that a promising direction for model adaptation is to provide video summary as its context, even without model fine-tuning.

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